Landing on the Wrong Altimeter Setting: A Pre-registered Study of Business Jets and Airline Jets in Open ADS-B Data

James Le Cuirot*
CitadelAero

Working draft v0.1, 11 November 2026

Abstract

An aircraft that lands with the wrong altimeter setting flies a barometric approach at the wrong height, and its crew has no indication of it. Two European studies have measured how often this happens; none has measured it in the United States or compared types of operator. Modern aircraft broadcast the altimeter setting the crew has selected. Using an open archive of these broadcasts, this pre-registered study measured the setting at landing for 713,835 landings at 119 US and European airports on 30 days in 2026, against the consensus setting of other aircraft landing at the same airport within 30 minutes. The setting differed by 2 hPa or more on 2.06 per 1,000 landings, by 5 hPa or more on 0.26 per 1,000, and by 10 hPa or more (about 280 ft) on 0.08 per 1,000. Business jets were wrong far more often than airline jets at the same airport on the same day: 8.5 against 1.4 per 1,000 at 2 hPa (Mantel-Haenszel odds ratio 3.4, 95% interval 2.9 to 4.0), and the gap widened with the size of the error (odds ratio 7.6 at 5 hPa and 15.8 at 10 hPa). The second hypothesis, that landing with the standard setting still selected is more common in Europe than in the United States, was not supported at the pre-registered threshold (rate ratio 2.0, $p = .07$). Large errors were about as frequent in the United States as in Europe. The reference is inferred from other aircraft, the broadcast is optional in the United States, and a third of business-jet landings could not be assessed, so the business-jet rates describe the better-equipped part of that fleet at busier airports.

1 Introduction

On 23 May 2022 an Airbus A320 on a satellite-based approach to Paris-Charles de Gaulle descended to about six feet above the ground almost a mile short of the runway. The controller had passed an altimeter setting 10 hPa too high, the crew set it, and every barometric indication on board looked normal [**?**]. The European Union Aviation Safety Agency has since issued and revised a safety bulletin on incorrect altimeter settings [**?**].

How often aircraft land on a wrong setting has been measured twice, both times in Europe. The French civil aviation safety directorate compared the setting downlinked to radar with the setting displayed to controllers for 1,694,266 landings at six French airports from 2019 to 2023. It found a difference of 2 hPa or more at the runway threshold on 2,269 landings (about 1.3 per 1,000) and of 10 hPa or more on 387 (about 0.23 per 1,000). The rate at Le Bourget, a business-aviation airport, was three times the rate at Charles de Gaulle, which the report attributed to "very different fleets" without further analysis [**?**]. A EUROCONTROL study built a detection algorithm on broadcast data and, as reported by France to an ICAO regional group, found 196 errors above 5 hPa in 747,353 flights at 378 European airports [**?**, **?**]. The full text of that paper could not be obtained for this study.

*Commercial pilot (Senior First Officer), aviation flight safety officer, and aviation safety and compliance consultant specialising in human factors; MSc in Human Factors in Aviation.

No published rate was found for the United States, and no study was found that compares business aviation with airlines or one region with another. The United States differs from Europe in two relevant ways. Settings are given in inches of mercury, and the change from the standard setting to the local one is made at 18,000 ft, well before the approach, where in Europe it is made at a few thousand feet [**?**, **?**].

This study uses open data to measure the rate at 119 airports in both regions and to test two pre-registered hypotheses: that business jets land on a wrong setting more often than airline jets at the same airport on the same day (H1), and that landing with the standard setting still selected is more frequent at European than at US airports (H2).

2 Methods

The hypotheses, measures and analysis were committed to a public repository before any altimeter-setting data from the study days was read (branch `prereg/08-wrong-altimeter-setting`, commit `be41aa0`, 11 November 2026, 13:18 UTC). Section **??** lists departures from that plan.

Data. The adsb.lol archive publishes each day's ADS-B traces under an open licence [**?**]. The study uses 30 UTC days in 2026, drawn at random for an earlier study of approach paths that never used the altimeter-setting field. Airports (119, chosen in advance, 76 in the United States and 43 in Europe) and aircraft types (65 business-jet and 58 airline jet types) were fixed lists. Airline-type aircraft were kept only with an airline-format callsign. Airport elevations came from OurAirports [**?**].

Setting at landing. A landing is the last airborne point of a track aligned with a runway and followed by ground movement or no later climb. The setting at landing is the last value of the selected barometric pressure setting broadcast by the aircraft in the 120 seconds before that point. The field has a resolution of 0.8 hPa [**?**, **?**]. Values outside 940 to 1060 hPa were dropped.

Reference. The reference for a landing is the median setting at landing of other aircraft landing at the same airport within 30 minutes either side. It required at least five other aircraft, at least 60% of them within 1.0 hPa of their median. Landings without a reference were excluded.

Errors. The difference is the setting at landing minus the reference. Thresholds were 2 hPa (primary), 5 hPa and 10 hPa. With 0.8 hPa steps, 2 hPa in practice means 2.4 hPa. "Standard left set" means a setting within 0.5 hPa of 1013.25 that differed from the reference by 2 hPa or more. Two exclusions were made without regard to class: settings matching station pressure at airports above 300 ft (use of QFE), and all landings by any aircraft that was wrong on at least half of four or more landings (suspect avionics).

Analysis. H1: Mantel-Haenszel common odds ratio of an error of 2 hPa or more, business jet against airline jet, across strata of airport and UTC date, with a Cochran-Mantel-Haenszel test, a Robins-Breslow-Greenland interval and an interval from resampling whole airports. H2: the ratio of the rate of "standard left set" at European airports to that at US airports, among landings where the reference was at least 3 hPa from standard, with a p-value from permuting the region label across airports and an interval from resampling airports. Holm correction across the two tests. The method was built on one pilot day outside the sample, using pooled results and a run with shuffled labels; no result split by class or region was computed until all 30 days were processed.

3 Deviations from the pre-registration

The frozen code was run unchanged and its hashes match the pre-registration. Section **??** reports analyses designed after the results were seen; they are labelled post hoc. The pre-registration counted 119 airports; 117 contributed landings to H2 because two had no eligible landing.

4 Results

4.1 Sample and coverage

The 30 days held 809,959 landings: 714,916 by airline jets and 95,043 by business jets. A setting was broadcast in the last two minutes on 97.0% of them. Coverage differed by group (Table **??**). The largest loss was among business jets, mostly because too few other aircraft landed within the hour to form a reference. After exclusions (27 implausible values, 3 landings on QFE, and 89 landings by 9 aircraft with suspect avionics), 713,835 landings were analysed: 650,307 by 17,617 airline aircraft and 63,528 by 11,618 business jets.

Table 1: Coverage by class and region.

Group	Landings	Setting broadcast	Analysed
Airline jets, Europe	275,237	99.6%	93.6%
Airline jets, United States	439,679	97.2%	89.3%
Business jets, Europe	17,416	97.9%	74.9%
Business jets, United States	77,627	86.9%	65.0%

4.2 How often

Across all analysed landings the setting differed from the reference by 2 hPa or more on 1,472 (2.06 per 1,000; 95% interval 1.88 to 2.26), by 5 hPa or more on 186 (0.26 per 1,000; 0.22 to 0.30), and by 10 hPa or more on 59 (0.083 per 1,000; 0.062 to 0.103). Table **??** gives the rates by region and class.

Table 2: Landings per 1,000 with the setting wrong by at least the amount shown.

	Landings	2 hPa	5 hPa	10 hPa
Europe, all	270,680	0.54	0.30	0.092
United States, all	443,155	2.99	0.24	0.077
Airline jets, Europe	257,629	0.47	0.24	0.066
Airline jets, United States	392,678	2.06	0.15	0.038
Business jets, Europe	13,051	1.76	1.38	0.61
Business jets, United States	50,477	10.28	0.95	0.38

The European rate above 5 hPa was 0.29 per 1,000 (0.22 to 0.37), close to the 0.26 per 1,000 reported for the EUROCONTROL study. The European rates at 2 hPa and 10 hPa (0.54 and 0.09) were below the French figures (about 1.3 and 0.23). Large errors were about as frequent in the United States as in Europe. Small ones were five times as frequent in the

United States; most of these were exactly three broadcast steps (2.4 hPa) and they were more common when pressure was changing.

4.3 H1: business jets against airline jets

Business jets landed on a setting wrong by 2 hPa or more on 8.53 per 1,000 landings (542 of 63,528) against 1.43 per 1,000 for airline jets (930 of 650,307). Within the same airport and date the odds ratio was 3.42 (95% interval 2.94 to 3.98; resampling airports, 2.77 to 4.27; Holm-adjusted $p < .001$; 684 informative strata). H1 is supported.

The difference grew with the size of the error (Figure **??**): odds ratio 4.8 (3.8 to 6.0) at 3 hPa, 7.6 (4.8 to 11.8) at 5 hPa and 15.8 (7.5 to 33.5) at 10 hPa. At 10 hPa the crude rates were 0.43 per 1,000 for business jets (27 landings) and 0.049 for airline jets (32 landings).

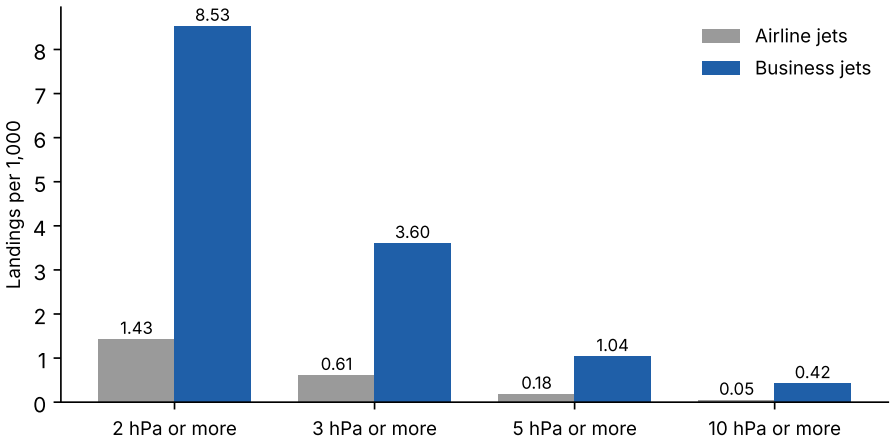


Figure 1: Landings per 1,000 with the altimeter setting wrong by at least the amount shown, by class. Crude rates over all analysed landings.

4.4 H2: standard setting left selected, Europe against the United States

Among landings where the reference was at least 3 hPa from standard, the standard setting was still selected on 71 of 188,811 landings in Europe (0.38 per 1,000) and 58 of 303,311 in the United States (0.19 per 1,000). The rate ratio was 1.97 (0.99 to 3.49; permutation $p = .072$). H2 is not supported at the pre-registered threshold.

5 Sensitivity, exploratory and post hoc analyses

Nothing in this section is confirmatory.

5.1 Pre-registered sensitivity analyses

The H1 odds ratio stayed between 3.1 and 5.1 under every pre-registered variation: excluding periods when the reference changed by 1.5 hPa or more in an hour (4.0), a 15-minute reference window (3.5), including the suspect aircraft (3.8), aircraft with three or more landings (3.1), the United States alone (3.3), Europe alone (5.1; 2.9 to 9.0), adding the hour to the strata (3.6), ground-confirmed landings only (3.2), and airport-only strata (3.4). Leaving out each airport in turn moved it between 3.29 and 3.68.

For H2, the result differed by class. Among airline jets the standard setting was left selected on 0.32 per 1,000 eligible landings in Europe against 0.074 in the United States (rate ratio 4.3; 1.8 to 9.2; $p = .005$). Among business jets the rates were 1.5 and 1.1 per 1,000 (ratio 1.4; $p = .42$). Requiring the reference to be 5 hPa from standard gave a pooled ratio of 3.2 (1.5 to 6.5; $p = .006$).

5.2 Pre-registered exploratory analyses

- **Standard left set, by class.** Business jets had the standard setting selected at landing far more often than airline jets at the same airport and date (odds ratio 13.1; 7.7 to 22.4). It accounted for 94 of the 145 errors of 2 hPa or more in Europe and 152 of 1,327 in the United States.
- **Direction.** Errors were evenly split: 700 too high, which puts the aircraft lower than indicated, and 772 too low.
- **Characteristic slips.** Differences of about 10 hPa occurred on 12 landings in Europe and 13 in the United States; differences of about one inch of mercury on 2 landings in the United States.
- **Already wrong five minutes out.** Of errors present at landing, 90% for airline jets and 97% for business jets were already present between ten and five minutes earlier. Airline jets that were wrong at that earlier point had corrected it by landing in 63% of cases, business jets in 15%. In Europe an aircraft may properly still be on the standard setting at the earlier point, so the airline correction figure is overstated.
- **Changing pressure.** When the reference had changed by 1.5 hPa or more in the previous hour, the rate at 2 hPa was 4.2 per 1,000 for airline jets against 1.2 otherwise, and 15.0 against 8.0 for business jets.
- **Within business jets.** Rates at 2 hPa per 1,000 were 7.4 for large jets on company callsigns, 6.6 for light jets on company callsigns, 9.9 for large jets on registrations and 13.6 for light jets on registrations. Large jets on company callsigns against airline jets gave an odds ratio of 2.5 (2.0 to 3.2).
- **Aircraft type.** Every business-jet type with at least 1,000 landings had a rate between 3.1 and 16.2 per 1,000. Airline types with at least 5,000 landings ranged from 0.1 to 5.1, with regional jets highest.
- **Day and night.** The H1 odds ratio was 3.4 by day and 3.6 at night.
- **Paris.** Le Bourget had 3 errors in 873 landings, Charles de Gaulle 30 in 13,831 and Orly 7 in 7,810. The sample is too small to confirm or refute the French threefold difference.

5.3 Post hoc analyses

A stuck or misreporting transponder would produce repeated errors by the same aircraft. The 542 business-jet errors came from 516 aircraft and the 930 airline errors from 853, so repeated errors were uncommon in both classes. The suspect-avionics rule can only act on aircraft with four or more landings, which favours airline aircraft. Restricting both classes to aircraft with four or more landings gave an H1 odds ratio of 3.1 (2.6 to 3.7) at 2 hPa and 5.5 (3.3 to 9.3) at 5 hPa. Counted per aircraft instead of per landing, 4.4% of business jets and 4.8% of airline aircraft had at least one error in the sample, but the airline aircraft made on average 37 landings each and the business jets 5.5.

6 Discussion

What the study shows. Landing on a wrong altimeter setting is rare, and it is several times more common in business jets than in airline jets using the same airport on the same day. The measure is direct: it is the value the aircraft reports as selected, compared with what other crews had selected at that place and time. The difference is present in both regions, by day and night, in every business-jet type with enough landings, and among large-cabin jets on company callsigns. It becomes larger, not smaller, as the threshold rises, which argues against an explanation based on small lags or rounding. This is consistent with the unexplained threefold difference between Le Bourget and Charles de Gaulle in the French data [**?**].

What kind of error. For business jets a large share of the gap is the standard setting still selected at landing, thirteen times as likely as for airline jets, and business jets that were wrong five minutes before landing rarely corrected it. Both point to a missed step at the change from standard to local setting that no later check catches, more than to mis-set digits. The study cannot say why. Candidate reasons include single-pilot or less formal cross-checking, fewer automated alerts on a setting mismatch, and approaches that do not depend on barometric guidance.

The United States against Europe. Large errors were about as frequent in the United States as in Europe, which does not support the assumption that the higher US transition altitude makes the problem a European one. Among airline jets, leaving the standard setting selected was four times as frequent in Europe, as the transition-altitude argument predicts, but this was an unplanned split of a test that did not reach significance overall. Small differences of two to three hectopascals were much more common in the United States. They rose with changing pressure and probably reflect settings that were correct when set and not updated.

Consequences. A 10 hPa error is about 280 ft, and errors in the unsafe direction were as common as in the safe one. On an approach with ground-based vertical guidance the error changes little. On an approach using barometric vertical guidance, or any non-precision approach, the aircraft follows a path displaced by that amount [**?**]. The study does not know which type of approach was flown.

7 Limitations

- **Inferred reference.** The reference is the consensus of other aircraft. A wrong value given to every aircraft would not be detected, and no weather reports were used to check it.
- **Coverage and selection.** Only 65% of US and 75% of European business-jet landings could be analysed, against about 90% for airline jets. The missing landings are aircraft that do not broadcast a setting, which is optional in the United States, and landings at times with too little other traffic. The business-jet result therefore describes better-equipped aircraft at busier airports and times, and may not hold elsewhere.
- **What the field represents.** The broadcast value is the setting selected on one side of the flight deck; which side is not defined, and some avionics may report stale values [**?**]. The exclusion rule for suspect avionics needs four landings and so screens airline aircraft better than business jets.
- **Lag counted as error.** A crew holding a correct but older value while pressure changes is counted. This mainly affects the 2 hPa threshold.

- **Resolution.** With 0.8 hPa steps the 2 hPa threshold is effectively 2.4 hPa.
- **No operational context.** Approach type, weather, minima, crew composition and what ATC said are unknown. Operator type is inferred from aircraft type and callsign.
- **Sample.** Thirty days in 2026, none in July, at 119 airports chosen by judgement.
- **Prior work not fully read.** The EUROCONTROL paper was seen only as an abstract and through secondary reports, so overlap with its breakdowns cannot be ruled out.

Acknowledgement

AI assistance. Literature searches, drafting and source verification were carried out with AI assistance (Claude, Anthropic) under the author's direction, as were the code and analysis. An independent AI-assisted audit checked the analysis and the claims against the outputs, and its findings were addressed before this version. The author is responsible for the final text.

Competing interests. The author is the founder of CitadelAero, which builds AI tools for aviation, and is employed as a business-jet pilot.

Disclaimer. The views expressed are the author's own and do not represent those of any employer or regulator. Nothing here is operational, legal or regulatory advice.

Materials. The pre-registration, code, results and audit record for this paper are at `https://github.com/JamesLeCuirot/Research/tree/main/papers/08-wrong-altimeter-setting`.